COSC2148/COSC3130 Computing Research and Project Preparation

Assignment 2

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Section 1

Introduction

1.1 DELETE original question

Introduction - 1 page (2 marks)

Briefly describe your topic and why it is important or interesting. You could start by describing the area of interest and later summarise the existing solutions and their limitations. You could also explain the specific focus of your research and why you are doing so. Feel free to include specific information, such as detailed statistics, that will back up your claims (and thus make your statements much stronger).

In summary, an introduction should:

- Introduce the topic and its significance.
- Give background and context.
- Outline your problem and research questions.

Some important questions to guide your introduction include:

- Who is interested in the topic (e.g. scientists, practitioners, policymakers, particular members of society)?
- How much is already known about the problem?
- What is missing from current body knowledge?
- What new insights will your research contribute?
- Why is this research worth doing

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1.3 Research Questions

This research aims to develop a context-aware misinformation detection framework that substitutes the text-only content branch used in established context-aware ensemble models, such as DANES **?** and GETAE **?**, with a multimodal content branch that jointly captures textual and visual information. While recent work has shown that multimodal content features and social context features each independently improve fake news detection, the closest existing combinations **???** use social branches that differ architecturally from the propagation-graph ensembles of the DANES/GETAE family. As a result, two questions remain unanswered, and together they form the technical core of this project.

RQ1. *How does replacing a text-only content branch with a multimodal content branch affect the performance of context-aware inauthentic social media content detection models, and to what extent?*

What it addresses. This question isolates a single architectural change: keeping the social context branch of DANES and GETAE intact while substituting the text-only content processor with a content branch that fuses textual and visual features (e.g., BERT for text and a CNN- or transformer-based visual encoder for images). The effect is measured against the original text-only baselines on the same benchmark datasets used in the DANES and GETAE papers (Twitter15 and Twitter16), as well as the multimodal Fakeddit dataset **?**, using accuracy, F1-score, precision, and recall.

Significance. Misinformation on social platforms increasingly relies on manipulated images, memes, and screenshots, where the textual signal alone is insufficient or misleading **?**. Even strong context-aware ensembles inherit the blindness of their text-only content component, meaning a measurable portion of detectable misinformation may be missed not because the social signal is weak, but because the content signal is incomplete. Quantifying the gain (or absence of gain) from a multimodal substitution directly informs whether the established DANES/GETAE family of architectures is leaving accuracy on the table.

Novelty and complexity. No prior work directly performs this controlled substitution and isolates its effect on the DANES/GETAE backbone. Existing multimodal-plus-context models such as EFND **?**, DPSG **?**, and the framework of Dellys et al. **?** use fundamentally different social branches (engagement signals, dynamic propagation graphs, or attention-based fusion), so their results cannot be cleanly attributed to the multimodal component. Answering RQ1 requires re-implementing a non-trivial baseline, designing a compatible multimodal branch, and ensuring the comparison remains controlled — a challenging but well-defined contribution.

RQ2. *How does the relative contribution of the multimodal content branch and the social context branch affect overall detection performance?*

What it addresses. This question moves beyond headline accuracy to decompose where the model's predictive power comes from. Through ablation studies (content-only, social-only, and full ensemble configurations) and analysis of the fusion-layer weights, the project will quantify how much each branch contributes to correct classification, and whether the two branches are complementary or redundant on different categories of misinformation.

Significance. Knowing that a combined model performs well is not enough; researchers and practitioners need to know *which* signal is doing the work. If the social context branch dominates, then the additional engineering cost of multimodal feature extraction may not be justified for deployment. Conversely, if the multimodal branch dominates, the field should reconsider its current emphasis on increasingly complex propagation-graph architectures. The answer therefore has direct design implications for the next generation of misinformation detectors.

Novelty and complexity. Most prior multimodal-plus-context studies report only aggregate performance and do not cleanly decompose contributions across a propagation-graph social branch of the DANES/GETAE family. Approaches such as TMEF-BI **?** perform reliability-aware weighting but on a different social signal (speaker behaviour rather than propagation), so their findings do not transfer directly. A principled contribution analysis on the proposed backbone is therefore both novel and methodologically demanding, requiring careful experimental design to avoid confounding the two branches.

Together, RQ1 and RQ2 form a coherent investigation: RQ1 establishes *whether* multimodal substitution improves context-aware misinformation detection, while RQ2 explains *why*, by revealing how the content and social signals interact. Answering both is necessary to make a defensible contribution to the context-aware fake news detection literature.

Section 2

Literature Review

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Literature Review - 2 pages (7 marks)

It is indeed essential to demonstrate that you are familiar with the most important research on your selected topic. A strong literature review convinces the reader that your research/project has a solid foundation in existing knowledge or theory. It also shows that you are not simply repeating what others have already done or said.

The aim of this section is to show exactly how your project will contribute to discussions in the field.

- Compare and contrast: what are the main theories, methods, debates, and controversies?
- Be critical: what are the strengths and weaknesses of different approaches?
- Show how your research fits in: how will you build on, challenge, or synthesise the work of others?

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Section 3

Research Methodologies

3.1 Research Design

Building on the existing DANES and GETAE ensemble architectures for inauthentic content detection, the proposed MACE (**M**ultimodal **A**nd **C**ontext-aware **E**nsemble) framework adopts a comparative experimental design to evaluate the impact of integrating multimodal content features into existing ensemble models.

3.2 System Architecture

DANES and GETAE are ensemble approaches fusing a text-based content branch and a social context branch to classify inauthentic content on social media.

HMCAN(Hierarchical Multi-modal Contextual Attention Network Qian et al. [2021]) and MPFN (Multimodal Progressive Fusion Network Jing et al. [2023]) are multimodal architectures performing inauthentic content detection based on textual and visual features in social media posts.

The proposed MACE framework retains the dual-branch structure and social context features of DANES and GETAE and replaces the text-based content branch with a multimodal branch (HMCAN or MPFN[1]) content branch, allowing for evaulation of the classification performance of multimodal content features within existing ensemble frameworks.

Diagram Here (TODO: insert architecture diagram showing the proposed modifications to the baseline architecture).

3.2.1 Multimodal Content Branch

MACE adopts HMCAN as the multimodal content branch, which is selected for its task alignment, dataset compatibility and ease of integration with the existing DANES/GETAE architectures.

[1] If HMCAN proves incompatible with the selected dataset or integration constraints, MPFN is identified as a potential fallback candidate, utilising a more complex image pipeline.

HMCAN has demonstrated excellent accuracy (89.7%, outperforming common baselines such as EANN) in multimodal fake news detection using its BERT + ResNet encoding pipeline, is consistent with the text branch of DANES/GETAE and has been independently validated on the Multimodal Twitter dataset, making it a suitable candidate.

3.2.2 Social Context Branch

We retain the Graph-based Deep Learning social context branches from the baseline DANES/GETAE architectures.

FOR DISCUSSION: Should we create a RQ to find out if these ensemble architectures can be adapted to a different dataset? we might find that this is not possible without significant modification, and this would have implications for the rest of the study. (This would be an interesting finding in itself), perhaps evaluating different datasets with these models might be an alternate research path? TK

	DANES	**GETAE**
Social context approach	Graph embedding from static network structure	GNN over dynamic propagation graph
Key information used	User interactions, relationships, and post behaviour	Information diffusion and propagation patterns

Table 3.1: Comparison of DANES and GETAE social context branches.

3.2.3 Feature Fusion and Classification

Both fusion models combine branch outputs by concatenating the content and context embeddings to produce a classification embedding, which is passed to a classification layer. The classification layer applies a softmax function to produce a probability distribution over the classes (authentic or inauthentic) and assigns a label based on the highest probability.

MACE retains this fusion mechanism to ensure the performance differences can be attributed to the substitution of the content branch rather than changes in the fusion or classification approach.

3.3 Experimental Design

The experiment design is structured as a two-stage process to systematically evaluate the proposed architecture and address the research questions. Both base architectures are evaluated allowing the findings to be evaluated for consistency across different fusion approaches.

The model is treated as a replication factor, rather than a variable, as the primary focus is on evaluating the impact of the proposed architecture modifications (multimodal content branch), rather than comparing the performance of the two fusion approaches.

3.3.1 Experiment Stages and Conditions

Stage 1 is to replicate the published architectures and evaluate their performance on the Twitter15 and Twitter16 datasets to ensure the implementation is correct and consistent with the published results.

Stage 2 evaluates four conditions on a multimodal dataset. Condition A establishes a new baseline and Conditions B, C and D systematically evaluate the contribution of content and context features against this new baseline to isolate the impact of the proposed architecture modification.

Stage		Label	Content Branch	Context Branch	Goal / Purpose
1	A	Implementation Baseline	Text-only	✓	Replicate baseline fusion models and confirm published results prior to introducing modifications.
2	A	Dataset Baseline	Text-only	✓	Establish baseline on multimodal dataset using existing fusion models.
	B	Ablation 1	Text-only	×	Isolate contribution of text-only features.
	C	Ablation 2	Multimodal	×	Isolate contribution of multimodal features.
	D	Proposed (MACE)	Multimodal	✓	Evaluation of full proposed framework.

Table 3.2: Summary of experiment conditions and objectives.

3.3.2 Research Question Alignment

RQ1 is directly addressed by comparing the performance of the proposed architecture (Condition D) against the baseline architecture (Condition A) with the multimodal dataset.

A pairwise comparison of condition's B, C and D address RQ2 by evaluating the contribution of the multimodal content branch and the social context branch to the overall performance of the model.

	Comparison	What varies	What it tells us
RQ1	A vs D	Content branch	Does substituting the multimodal branch for the text branch improve performance within the full context-aware framework?
RQ2	A vs B	Context branch	How much does the social context branch contribute with text-only content?
	B vs C	Content branch	Does adding visual features improve detection independent of social context?
	C vs D	Context branch	How much does the social context branch contribute with multimodal content?

Table 3.3: Pairwise condition comparisons and research question alignment.

3.4 Project Timeline

Section 4

Evaluation

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evaluation - 1 page (3 marks)

This section is important and should clearly show the evaluation plan of your research. This evaluation plan could be used later during your thesis to clearly demonstrate that you made research contributions in the field of study.

This section will need to address the following points:

- Benchmarking: clearly select some existing methods/algorithms you will need to compare your research work with. It is important to clearly explain these methods/algorithms as well as justify the reasons for choosing them.

- Data sets: describe at least two well-accepted data sets that you will be using to compare your research work against the selected benchmarked methods/algorithms. Again, you will need to justify the reasons for selecting such data sets.

- Metrics: to compare your research work with benchmarked methods/algorithms, you clearly need to list the metrics to be used for this purpose. This could be, for example, "execution time" if you are comparing the performance of your work with other methods/algorithms. You also need to explain why you have selected such metrics. List at least two metrics.

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TIMS CONTENT - I ACCIDENTALLY WROTE THIS IN THE METHODOLOGY CHAPTER BUT THEN I REALISED I DIDN'T NEED IT, SO INCORPORATE IF NEEDED I GUESS?

4.3 Datasets and Data Preparation

Twitter15 and Twitter16 are used for Stage 1 replication only, as they do not contain image data.

For Stage 2, a multimodal dataset will be used containing multimodal content (text and images) and social context features through a conversation thread structure, comparable to Twitter15/16 (to ensure the social context branch can be retained with minimal modification).

We are not intending to derive conclusions about the relative performance of the models on the Twitter15/16 datasets and the multimodal dataset so comparisons between stage 1 and 2 are not relevant[1].

TODO this needs fleshing out a bit more I think

4.4 Evaluation Strategy

The four-condition design allows for systematic evaluation of the proposed architecture modification by isolating the impact of multimodal content and social context features through controlled ablation experiments.

Performance is evaluated using accuracy, precision, recall, and F1-score, which are standard metrics for classification tasks and allow for comprehensive assessment of model performance across both classes (authentic and inauthentic).

ROC AUC will also be used to evaluate the model's ability to distinguish between classes across different thresholds.

[1] It is anticipated that the transition from Twitter15/16 datasets to a multimodal datasets will result in a performance difference in Stage 2 Condition A relative to Stage 1 depending on the extent to which these models generalise to a new dataset.

Section 5

Project Management

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project management - 1 page (1 mark)

This section should outline how you plan to manage your research project throughout the semester.

It is essentially a Project Plan that clearly defines the scope and deliverables, including explicit exclusions.

It should present a structured schedule with milestones and a transparent critical path, while assigning responsibility to specific 2individuals to ensure accountability.

It must also incorporate risk management to anticipate threats and formulate appropriate responses, protecting project objectives from avoidable failure.

This section should also clearly state the methods you will use to manage the project, such as Agile sprints, Scrum ceremonies, Kanban, etc., and specify the enabling tools (e.g., MS Project, Trello, JIRA, Slack, etc.) that will operationalise these methods.

Briefly explain how these methods and the supporting tools together will facilitate scheduling, progress tracking, communication, and reporting. Specifically, you must include the following:

- High-level scope and deliverables of your research project. Clearly state exactly what you will deliver and explicitly rule out what you will not.
- Timeline with major milestones derived from the Work Breakdown Structure (WBS). This should reflect the schedule of your research project and include any critical paths. Note: you don't need to include the WBS itself.
- Clear assignment (ownership) of each task to project team member(s), not roles, for accountability and monitoring.
- Approach in risk management, identify risks and proposed mitigation and contingency plan, as well as escalation paths. List the major risks, rank them, and state what you will do before, when, and after they occur.

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Bibliography

J. Jing, H. Wu, J. Sun, X. Fang, and H. Zhang. Multimodal fake news detection via progressive fusion networks. *Information Processing & Management*, 60(1):103120, 2023. ISSN 0306-4573. doi: https://doi.org/10.1016/j.ipm.2022.103120. URL https://www.sciencedirect.com/science/article/pii/S0306457322002217.

S. Qian, J. Wang, J. Hu, Q. Fang, and C. Xu. Hierarchical multi-modal contextual attention network for fake news detection. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '21, pages 153–162, New York, NY, USA, 2021. Association for Computing Machinery. ISBN 9781450380379. doi: 10.1145/3404835.3462871. URL https://doi.org/10.1145/3404835.3462871.

Appendix A

Generative AI Usage

Declaration

[**REPLACE THIS LINE WITH YOUR GENERATED DECLARATION FROM** `https://aiattribution.github.io/`]

Tool Report: [Tool Name, e.g. ChatGPT / RMIT Val / Copilot]

Field	Details
1. GenAI tool used	[e.g. ChatGPT 4o, RMIT Val, GitHub Copilot]
2. Purpose for this assignment	[e.g. Brainstorming research questions, drafting outline, rewriting for clarity, code generation]
3. Prompts or instructions given	[Include key prompts used. If conversation was lengthy, include all your prompts in full and use [. . .] to truncate AI responses only.]
4. Number of prompts given	[e.g. 7]
5. Summary of GenAI output	[Describe what the AI produced. Include brief excerpts where relevant.]
6. Verification of GenAI output	[Explain how you checked the AI's output for factual accuracy, source validity, and logical soundness. e.g. "All cited sources were manually located and verified via Google Scholar."]
7. Usage or modification of output	[Explain how you used, adapted, or discarded the AI-generated content in your final submission.]
8. Reflection	[What worked well? What did not? How did it support your learning? What value did it add?]

Table A.1: Generative AI usage report — [Tool Name]